VIGIL: Validity-Gated Retrieval for Product–Litigation Risk Entailment

Anonymous ACL submission

Abstract

Retrieval-augmented generation (RAG) treats semantically similar documents as interchangeable evidence, but in law a lexically close precedent may already have been narrowed, superseded, or overruled at the date that matters. We study this failure in a product-facing setting and introduce *Product–Litigation Risk Entailment* (PLRE), a directional task that asks whether an AI product's attributes entail exposure to a legal-risk theory under precedent that is valid at a given query date. Our system, VIGIL (Validity-Interval Gating in Litigation), infers *when* each authority stops being operative: it extracts treatment edges (overrules, supersedes, narrows) from later opinions, turns them into validity intervals, and validates them against gold citation spans (negative-treatment micro-F1 0.80) and a commercial citator (micro-F1 0.81). A deliberately simple retrieval-time gate applies these intervals before generation, and a goal-conditioned agentic loop stops when evidence coverage, not legal correctness, is satisfied. On VIGIL-Dataset (1,247 U.S. AI-litigation cases, 312 product-mapping cases), validity gating raises held-out PLRE F1 from 0.72 to 0.78 and cuts the stale-citation rate from 12.7% to 4.1% relative to the same pipeline without the gate (point estimates). On a human-adjudicated triage subset (n=82) VIGIL reaches macro F1 0.79 vs. 0.75 for an iteration-matched baseline; that gap is secondary and not significance-tested. We release the dataset with per-record label provenance, annotation guidelines, and evaluation scripts. VIGIL is decision support for qualified reviewers, not legal advice.

1 Introduction

Generative-AI products face growing litigation over training data, output reproduction, model behavior, and downstream misuse (Henderson et al., 2023; Levendowski, 2018; Grimmelmann, 2016). High-profile suits such as *Getty Images v. Stability AI* (Getty Images, Inc., 2023), *New York Times v. Microsoft* (The New York Times Company, 2023), and *Kadrey v. Meta* (Kadrey et al., 2023) illustrate the range of theories from DMCA §1202 information removal to direct infringement and vicarious liability for outputs. For the engineering organizations that build these systems, the operational question is not merely whether a lawsuit exists, but whether a theory of liability maps onto a specific product attribute, whether the supporting authority remained valid at the relevant date, and which mitigation should be reviewed by counsel. Legal-operations teams track dockets and news, yet the bridge from a litigation event to a product-specific engineering action is usually manual. Keyword alerts surface noisy filings, docket trackers rarely map cases to product features, and conventional retrieval-augmented generation (RAG) (Lewis et al., 2020) treats semantically similar legal documents as interchangeable even when later decisions have narrowed or superseded them.

We formalize this need as *Product–Litigation Risk Entailment* (PLRE): deciding whether product attributes entail exposure to a legal-risk theory under time-valid precedent. PLRE differs from standard legal textual entailment (Rabelo et al., 2022) in three respects: the premise is an engineering description rather than legal text, the relation is directional (product attribute $\rightarrow$ risk theory), and the support set is conditioned on which authority was operative at the query date. Unlike commercial citators such as Shepard's and KeyCite, we construct a public-corpus, automatically extracted, time-indexed treatment layer and connect it directly to product-facing entailment.

We present VIGIL (**Validity-Interval Gating in Litigation**), a continuous sensing system built around this task. Its core component is a supersession-aware treatment layer that infers when each authority ceases to be operative from treatment edges in later opinions. A simple

retrieval-time gate (Eq. (3)) applies these intervals before generation, rather than flagging stale citations after an LLM has already produced a report. Around this layer, a goal-conditioned agentic loop exposes unmet evidence requirements and stops when a coverage threshold is met. Although the layer is defined for any time-stamped corpus in which later documents can supersede earlier ones, we evaluate it only for law within VIGIL.

Our contributions are:

- a supersession-aware treatment layer that extracts treatment edges from public opinions, validates them against gold spans and a commercial citator, and propagates them into validity intervals that gate retrieval at query time (§3.4);
- PLRE, an initial product-conditioned legal-risk entailment benchmark with non-VIGIL baselines and ablations that separate temporal validity from flat similarity (§3.5, §5);
- a goal-conditioned agentic loop with a coverage-based stopping rule; a conformal omission-risk bound holds under exchangeability and is reported as an auxiliary control (it is violated under temporal shift; §2.1, §5);
- VIGIL-Dataset with per-record provenance and a human-adjudicated subset (§4), plus a public SCOTUS overruling pair resource (Appendix H.1; external-stack pilot is incomplete).

2 Problem Formulation

Let $\mathcal{D} = \{d_1, \ldots, d_n\}$ be public litigation documents with metadata and text, and let P be a product specification describing architecture, data provenance, modality, and deployment context. VIGIL returns a structured risk report R^* and mitigation candidates G^* that must satisfy an auditable goal predicate $\mathcal{G}$ (§2.1). Report quality is not optimized directly; we evaluate it by severity-label F1 (RQ1), blinded report ratings (RQ2), and the coverage loss L_τ (RQ5). The core difficulty is a domain gap: product specifications employ engineering terms (data pipelines, API distribution), whereas litigation documents speak in causes of action, procedural posture, and precedent. PLRE (§3.5) therefore requires mapping engineering-language attributes onto legal-risk theories while conditioning the support set on authority that remains valid at the query date.

2.1 Goal Predicate and Stopping Rule

VIGIL treats report generation as goal-conditioned planning. The predicate is

$$\mathcal{G}(R) = \mathcal{G}_{\text{risk}}(R) \wedge \mathcal{G}_{\text{evidence}}(R) \wedge \mathcal{G}_{\text{guide}}(R), \quad (1)$$

where $\mathcal{G}_{\text{risk}}$ requires every risk item to carry a severity label, $\mathcal{G}_{\text{evidence}}$ requires at least k supporting documents above relevance threshold τ, and $\mathcal{G}_{\text{guide}}$ requires mitigation text for each High or Medium item. The controller selects among SEARCH, ANALYZE, SUMMARIZE, MAP, VERIFY, and REPORT; the deterministic (LLM-free) Goal Checker either terminates, emits a structured critique that closes residual gaps, or escalates unmet conditions. For example, if a High-risk item is supported by only one valid authority after gating, the critique records the missing theory, product attribute, jurisdiction, query date, and required evidence count; the Planner converts this critique into narrower precedent and entity queries rather than simply retrying the same prompt.

Let $\Phi(R_t)$ count the unmet atomic sub-conditions of $\mathcal{G}$, and let $\mathcal{T}_c^{\text{H}}$ be the human-labeled High-risk theories of case c that are *retrievable*—i.e., supported by at least one authority in $\mathcal{D}$ that is valid at t_q. The coverage loss at evidence threshold τ is

$$L_\tau(c) = \mathbf{1}\big[\exists\, T \in \mathcal{T}_c^{\text{H}} :\ T \notin R^*\big]. \quad (2)$$

Proposition 1 (Coverage-controlled stopping with escalation)**.** *Assume the set of risk items under consideration is fixed within a run (the Reporter may rewrite mitigation text but does not introduce new risk atoms that expand Φ). Under monotone evidence caching and attainable progress, $\Phi(R_t)$ is non-increasing, reaches zero in at most $\Phi(R_0)$ successful progress steps when all atoms are attainable, and otherwise returns the residual atoms at $T_{\max}$. If $\hat{\tau}$ is selected on exchangeable calibration cases by conformal risk control, then the next-case omission risk satisfies $\mathbb{E}[L_{\hat{\tau}}] \le \alpha$.*

The first clause follows almost directly from the definitions: monotone caching ensures that no satisfied atom becomes unmet, and attainable progress (each pass closes at least one attainable atom) is a design assumption that we do not verify directly. The second clause is a standard application of conformal risk control (Angelopoulos et al., 2024) to L_τ, which is monotone in τ (Appendix G). It governs evidence coverage only, holds only for exchangeable cases—our temporal split violates it at

$\alpha \in \{0.05, 0.10\}$ (§5)—and its tightness depends on the calibration-set size (150 cases).

3 Method: The VIGIL Framework

3.1 Overview

VIGIL comprises two layers. A sensing pipeline discovers AI-litigation filings in CourtListener/RECAP (Free Law Project, 2023) and legal news and converts them into case records; this engineering infrastructure is documented in Appendix C (architecture in Figure A.1). The remainder of this section describes the evaluated NLP components in order: LegalRisk-RAG (§3.2), the goal-conditioned agentic loop (§3.3), the temporal precedent graph (§3.4), and the PLRE decision procedure (§3.5).

3.2 LegalRisk-RAG

The LegalRisk-RAG pipeline adapts standard RAG to legal text in three respects. **Chunking** segments documents at logical structural boundaries—numbered claims, "WHEREFORE" clauses, section headings—rather than fixed token windows; each chunk retains a metadata header (case identifier, filing date, court level, section type). **Embedding**: we compared a dense retriever (Karpukhin et al., 2020) (`text-embedding-3-large`), RECAP-tuned LegalBERT (Chalkidis et al., 2020), and a hybrid BM25 + neural re-ranker on a 200-pair development set; the hybrid performed best and is the default (Table A.7, appendix). **Dynamic context expansion (precedent chaining)**: when a retrieved chunk references a prior case (e.g., *Authors Guild v. Google*), the pipeline appends that opinion as secondary context and queries the temporal precedent graph for subsequent treatment, so that the Analyzer receives both semantic neighbors and validity metadata (cited positively, narrowed, superseded, or overruled at the filing date).

3.3 Goal-Conditioned Agentic Loop

The four specialized agents and the central Goal Checker operate as in Algorithm 2. The **Planner** decomposes unmet sub-goals into ordered retrieval/analysis sub-tasks; the **Retriever** runs LegalRisk-RAG queries and reformulates via legal entity term expansion when $\mathcal{G}_{\text{evidence}}$ is unmet; the **Analyzer** computes $\text{RiskScore}(d_i, P, C)$, clusters evidence, and maps clusters to a risk taxonomy (Training Data, Output Reproduction, API Misuse, Model Weights); the **Goal Checker** evaluates $\mathcal{G}(R_t)$ after each pass and returns machine-readable residuals such as `missing(valid-authority, DMCA, output-CMI, NDCA, 2025-02)`; the Planner converts these critiques into expanded entity/date/theory queries; and the **Reporter** assembles the risk register, per-risk evidence with citations, and mitigation guidance.

3.4 Temporal Precedent Graph

VIGIL's central component is a treatment layer that determines *when* each authority ceases to be operative; a simple retrieval-time gate then applies it. The layer is realized as a temporal precedent graph $\mathcal{P}_t = (V, E_t)$. Each node $v \in V$ denotes a case, order, or doctrinal rule and stores jurisdiction, court level, issue tags, filing and decision dates, and a validity interval $[s_v, e_v)$, where e_v remains open unless the node is later vacated, overruled, or superseded; material narrowing instead lowers its treatment weight. Directed edges encode treatment relations: CITES, DISTINGUISHES, NARROWS, SUPERSEDES, and OVERRULES. Retrieval for a query at date t_q first obtains semantically relevant candidates and then applies validity gating:

$$\text{score}(v, q, t_q) = \text{sim}(v, q) \cdot \mathbf{1}[s_v \le t_q < e_v] \cdot w(\text{treat}(v, t_q)). \quad (3)$$

Here $w(\cdot)$ is a fixed monotone schedule: positively treated or still-open authorities (CITES with positive/neutral polarity, or no negative treatment) receive weight 1.0; *cautionary* authorities—those labeled DISTINGUISHES or bearing weak negative polarity without interval closure—receive 0.75; material NARROWS receives 0.50; fully closed (overruled or superseded) authorities are already excluded by the indicator. The operator separates legal authority from topical similarity: a stale but lexically close case may still appear as context, yet it cannot support a risk decision without passing the validity check. Equation (3) is therefore a constrained retrieval operator over (q, t_q): a host retriever scores semantic relevance as usual, and PLRE receives only support whose legal status remains operative at t_q. Unlike a learned temporal re-ranker (which can still rank closed authority highly) or a post-hoc citator warning applied after evidence is selected, the gate hard-filters support before generation; we leave a head-to-head re-ranker baseline to future work.

Why not a date filter? Once intervals are known, Eq. (3) is a metadata filter any vector store can

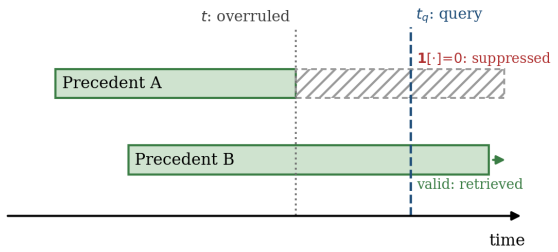


Figure 1: Validity gating (Eq. (3)). An overruling at t closes Precedent A, so a query at $t_q > t$ zeroes its indicator and suppresses it despite lexical similarity, while still-open Precedent B is retrieved—the time-valid distinction the RQ3 ablation isolates.

Table 1: Treatment-layer validation (neg.-treatment micro over OVERRULES/SUPERSEDES/material NARROWS).

Reference	P	R	F1
Gold spans (n=600)	0.84	0.77	0.80
Citator (Protocol P2)	0.83	0.80	0.81

apply. The difficulty is upstream and conceptual: e_v is set by *later* documents that overrule, supersede, or narrow v, so it is not recoverable from v's own filing date. A pure recency or document-date filter therefore cannot implement the same operator; the empirical question is whether treatment edges can be extracted reliably enough to use. Table 1 addresses that question: negative-treatment micro-F1 is 0.80 on 600 gold spans and 0.81 against a commercial citator (Protocol P2; missed-action rate 20.4%; 3.8% of held-out PLRE pairs flip after citator correction). Per-relation tables, error propagation, and a weaker regex baseline appear in Appendix D, H.2.

Graph construction. Nodes are induced from CourtListener/RECAP with intervals initialized to $[s_v, \infty)$. Edges are extracted by a citation-span detector plus an LLM classifier (GPT-4o, T=0; distinct from the controller's T=0.2); interval-closing edges must be anchored to a detected span, decision date, and source document, and gold evaluation uses spans held out from prompts. A confirmed OVERRULES/SUPERSEDES edge at t sets $e_v \leftarrow t$; material NARROWS sets w=0.50 from t. The graph has 1,812 nodes and 4,760 edges.

Low-confidence negative-treatment edges are routed to human review rather than auto-closing intervals; downstream effects of missed closures are audited in Table 6.

Table 2: Schematic PLRE instances (illustrative; not drawn from VIGIL-Dataset). The same attribute–theory pair can be positive or negative depending on whether its supporting authority is still operative at t_q.

Attribute p_i	**Theory** T_j	**Support at** t_q	y_{ij}
web-scale image scraping for training	training-data copyright infringement	operative authority on unlicensed training copies	1
same	same	only authority whose interval closed before t_q (stale distractor)	0
output filter removes copyright-management information	DMCA §1202 (output CMI)	no authority in the query's jurisdiction	0

3.5 PLRE Decision Procedure

We cast product–litigation alignment as a risk-entailment task. Given a product spec P with attributes $\{p_1, \ldots, p_m\}$ (training-data sources, collection method, output modality, architecture, deployment context, and capability claims), a legal-risk theory T_j, and a query date t_q, PLRE predicts:

$$y_{ij} = \mathbf{1}[P, p_i, \mathcal{P}_{t_q} \models T_j]. \quad (4)$$

Table 2 provides schematic instances. Unlike symmetric similarity, PLRE is directional: an attribute such as "web-scale image scraping" can support exposure to a training-data copyright theory, whereas the inverse relation is not meaningful. Unlike standard legal textual entailment, the premise incorporates product attributes and the entailment relation is conditioned on time-valid precedent. A pair (p_i, T_j) is promoted to the risk register only when (i) the entailment score exceeds the validation-set threshold, (ii) the top supporting precedents remain valid at t_q, and (iii) the report includes mitigation text grounded in at least three cited authorities when available.

PLRE gold labels are constructed from anonymized product-attribute cards and docket-grounded theory cards. Annotators judge a directional pair as positive only when the product attribute, legal theory, query date, and at least one valid supporting authority jointly establish exposure; negative pairs comprise ordinary mismatches and stale-authority distractors. Critically, the validity judgment encoded in the labels is produced by annotators reading docket and opinion text, *independently of VIGIL's automatically extracted treatment graph*, so the gate must recover label validity

rather than presuppose it. An extractor-independent re-check of these validity judgments against an authoritative citator forms the downstream link of Protocol P2. Product archetypes are tracked in the split manifest, and year and litigation type are stratified to reduce temporal and topical leakage.

The formulation is system-agnostic: PLRE is directional entailment subject to an exogenous, time-indexed support constraint, and any system can be evaluated on it. It is more demanding than ordinary legal NLI because a model must align engineering terms with legal predicates *and* reject stale yet lexically attractive precedent.

4 Dataset

VIGIL-Dataset is a curated benchmark of AI-related copyright and intellectual-property litigation designed for structured risk analysis. We describe its construction, statistics, and annotation protocol below; additional inclusion criteria and scoring details appear in Appendix A–B.

4.1 Construction and Statistics

The dataset compiles CourtListener/RECAP filings and supplementary legal-news signals spanning January 2020 to December 2025, with the majority of filings concentrated in 2023–2025. Cases were obtained via a retrospective keyword search of RECAP complaints (terms covering generative models, training data, model weights, and AI-assisted products) and were manually screened to confirm that AI development, training, or deployment was at issue. The scope is intentionally broader than copyright-only trackers: it includes privacy, publicity, product-liability, and contract theories when AI systems are material to the dispute, and counts each docket separately (including related and consolidated filings). One case corresponds to one docket; consolidated actions are therefore counted per docket (App. A.1). Table 3 summarizes the final annotated subset (1,247 dockets after screening and dedup; ≈980 distinct disputes). Annual post-screening counts by litigation type appear in Table A.1 (appendix).

Each record contains docket metadata, parties, document URLs, and a human-verified litigation summary. Status labels encode workflow state rather than adjudicated outcome; severity is reported separately and is not interpreted as liability.

Table 3: VIGIL-Dataset summary statistics. The three-class risk label merges *critical* and *high* into *High*. Status counts (initiated/active/settled/dismissed) sum to the full set of 1,247 cases.

Attribute	Value
Total cases	1,247
Total documents	8,934
Coverage period	2020–2025
Unique defendants	43
Risk labels	3 (High/Med/Low)
Status dist. (Init/Active/Settled/Dismissed)	298/695/148/106
Product-mapping GT cases	312

4.2 Annotation and Availability

An automated pre-labeler first assigns a triage score using the additive matrix in Table A.2. Three legal research specialists then independently review a stratified sample of 400 cases under a fixed guideline. During adjudication, annotators see docket text, product attributes, candidate theory cards, and query dates, but neither system identity nor the pre-labeler's score. The remaining 847 cases retain automated pre-labels; a 5% spot-check yields 91% agreement with at least one specialist (a lenient criterion). Inter-annotator agreement on the 400-case sample is mean pairwise Cohen's $\kappa = 0.74$ across the three annotator pairs (Landis and Koch, 1977) (Fleiss' $\kappa = 0.72$). Because retained labels can inflate metrics, we release per-record provenance, restrict all primary claims to the human-only evaluation, and treat retained-label results solely as scale-oriented comparisons. The top three defendants account for 41.3% of cases, limiting representativeness for smaller vendors and non-U.S. jurisdictions. The dataset, guidelines, split files, seeds, and evaluation scripts are supplied as anonymized supplementary material for review (App. F).

5 Experiments

We evaluate VIGIL along five research questions:

- **RQ1**: Does VIGIL improve severity-label agreement over strong baselines?
- **RQ2**: Do blinded specialists rate VIGIL reports more highly, especially on evidence sufficiency?
- **RQ3**: How much do the temporal gate and agentic loop each contribute?
- **RQ4**: Does validity gating improve Product–Litigation Risk Entailment?
- **RQ5**: Does the calibrated stopping rule control

evidence-omission risk under temporal shift?

5.1 Experimental Setup

Dataset split and label provenance. VIGIL-Dataset is partitioned into stratified train/validation/test splits by year and litigation type (not by dispute identity; related dockets of the same dispute may therefore appear in more than one split—a residual leakage risk we flag in Limitations). Because a substantial share of labels originate from an automated pre-labeler and only a subset are human-adjudicated, primary claims are computed exclusively on the human-adjudicated test split (n=82 cases); retained-label results are reported solely for scale. Conformal calibration draws on human-adjudicated cases from the training and validation pools, kept disjoint from the test cases. This design keeps the rule-based pre-labeler out of the evidence for our primary claims. For VIGIL and baselines B2–B5 the training split is used only to select few-shot examples and tune thresholds; no LLM is fine-tuned.

Baselines. We compare VIGIL against five risk-classification baselines: **B1 Additive-only Rule** (ingestion matrix without composite bonuses; thresholds tuned on validation); **B2 Single-Pass LLM**; **B3 RAG-only**; **B4 Single-Agent** (VIGIL without the goal-conditioned loop); and **B5 Iteration-Matched Self-Refinement** (a Reflexion-style baseline (Shinn et al., 2023) given VIGIL's mean budget of 2.4 iterations, matched on iterations rather than tokens). B2–B5 share the same GPT-4o backbone and decoding settings as VIGIL. For PLRE we further include non-VIGIL baselines that isolate suspected confounds: product–theory cosine, LegalBERT product–theory NLI, COLIEE-style (Rabelo et al., 2022) precedent retrieval without interval closure, and VIGIL without temporal gating. These baselines disentangle product–theory matching, legal entailment, retrieval quality, test-time compute, and the hard validity gate.

Automatic evaluation. All reported automatic metrics freeze the treatment graph produced by the extractor; low-confidence edges are *not* human-reviewed at evaluation time (review routing is a deployment option only). **Evaluation metrics.** We report macro F1/precision/recall over three severity labels, report-quality ratings by two blinded specialists, loop efficiency, PLRE precision/recall/F1, edge-error propagation, and evidence-coverage omission risk under calibrated thresholds. Severity-label F1 measures triage-label agreement, not legal liability. Calibrated omission risk and backbone robustness appear in the appendix; hyperparameters are listed in Table A.8.

Table 4: Macro severity-label agreement. Human-adjudicated subset (n=82; primary). Full: held-out test including retained pre-labels.

System	Human-adjudicated F1	P	R	Full F1
B1 Additive-only	0.55	0.55	0.56	0.62
B2 Single-Pass LLM	0.70	0.70	0.71	0.72
B3 RAG-only	0.72	0.72	0.72	0.74
B4 Single-Agent	0.74	0.74	0.74	0.76
B5 Self-Refinement	0.75	0.75	0.75	0.77
VIGIL Full	**0.79**	**0.79**	**0.80**	**0.81**

5.2 RQ1: Severity-Label Agreement

Table 4 reports severity-label agreement on the human-adjudicated subset (n=82). VIGIL reaches 0.79 macro F1 vs. 0.75 for iteration-matched B5; the same rank order holds on the full held-out test set. These triage numbers are secondary to PLRE and are not significance-tested. B1 omits the composite bonuses used by the pre-labeler, so its full-test F1 (0.62) is not circular.

5.3 RQ2: Report Quality

Two blinded specialists rated VIGIL and B2–B5 reports on 15 cases (75 reports) along Accuracy, Completeness, Actionability, Evidence Sufficiency, and Clarity (5-point scales). VIGIL mean scores are 4.3/4.1/4.2/4.4/3.9 on the five scales (Evidence Sufficiency highest). We do not claim pairwise superiority over B2–B5 without reporting their means here. Mean inter-rater $\kappa = 0.68$ (substantial agreement under Landis and Koch's scale; (Landis and Koch, 1977)). RQ2 is supporting evidence only (n=15).

5.4 RQ3: Agentic Loop and Temporal Graph Ablations

Temporal precedent reasoning is the central empirical claim. Table 5 isolates the two mechanisms of Eq. (3). Full validity-interval gating plus treatment weighting yields PLRE F1 of 0.78 and a stale-citation rate of 4.1%. Removing treatment weighting alone drops F1 by 0.03 and roughly doubles stale citations (7.8%). Removing interval gating entirely (recovering flat semantic similarity) widens the gap to ΔF1 $=-0.06$ and raises the stale-citation rate to 12.7%—a threefold increase. Flat

Table 5: Temporal precedent-graph ablation (main claim). Both mechanisms of Eq. (3) are removed in sequence; the bottom row recovers flat semantic similarity. Full gating cuts the stale-citation rate from 12.7% to 4.1% while improving PLRE F1 by 0.06. Stale-cite measures citations to already-closed precedent ($t_q \geq e_v$).

Variant	PLRE F1	ΔF1	Stale-cite %
VIGIL (Full: gating + weighting)	0.78	—	4.1
− treatment weighting $w(\cdot)$	0.75	−0.03	7.8
− interval gating (flat similarity)	0.72	−0.06	12.7

Table 6: Edge-error propagation audit. Gold row substitutes labels on the 600 annotated spans only. "Mask x%" rows also remove x% of extracted negative-treatment edges (closures or down-weights). Effective miss rate $\approx 1 - 0.77(1 - x)$ from the extractor's 0.77 recall; ΔF1 is relative to the extractor row.

Setting	Eff. miss	PLRE F1	ΔF1	Stale-cite %
Gold negative edges (on spans)	0%	0.80	+0.02	3.7
Extractor output	≈23%	0.78	—	4.1
+ mask 10% of extracted	≈31%	0.76	−0.02	6.2
+ mask 23% of extracted	≈41%	0.74	−0.04	8.9
+ mask 30% of extracted	≈46%	0.72	−0.06	11.3

retrieval systematically promotes lexically similar but already-closed authorities over still-operative ones, altering the evidence that reaches the generator. This controlled degradation constitutes the paper's main methodological result.

Table 6 measures how negative-treatment errors propagate. The "Gold negative edges" row substitutes gold-labeled negative-treatment relations on the 600 annotated citation spans into the live graph for those spans only (the remaining extracted edges are left unchanged); it is not a full replacement of all 4,760 graph edges. Residual stale-cite under gold (3.7%) arises when the generator cites a closed authority that entered context through precedent chaining rather than as a gated support document, or when annotator validity judgments disagree with interval closure on boundary cases. The extractor row operates at the observed miss rate (≈23%, from 0.77 recall) and costs 2 PLRE F1 points relative to that gold-span substitution. Rows below additionally mask a random fraction of the *extracted* negative-treatment edges before retrieval: removing a further 23% of them (≈41% effective miss rate) roughly doubles the stale-cite rate (4.1%→8.9%) and lowers PLRE F1 by 4 points.

Component ablations for the agentic loop confirm each part contributes: removing the Goal Checker, Planner, and Retriever query-expansion lowers full-test-set F1 from 0.81 to 0.76/0.78/0.79 respectively, and only query-expansion removal degrades retrieval (Recall@10 $0.84 \rightarrow 0.76$); details are in Table A.6.

Table 7: PLRE comparison against external legal-NLI and precedent-retrieval baselines. External systems do not use VIGIL's goal predicate or validity-interval closure. Held-out test pairs only.

System	Precision	Recall	F1
Product–theory cosine	0.64	0.66	0.65
LegalBERT product–theory NLI	0.69	0.68	0.68
COLIEE-style retrieval, no gate	0.71	0.70	0.70
RAG-only generation	0.73	0.74	0.73
VIGIL without temporal gating	0.72	0.73	0.72
VIGIL Full	**0.77**	**0.80**	**0.78**

5.5 RQ4: Product–Litigation Risk Entailment

PLRE uses anonymized product-attribute cards (five archetypes: training-data ingestion, output reproduction, API distribution, model-weight release, multimodal scraping) paired with docket-grounded theory cards across 312 product-mapping cases (1,856 pairs; 28.4% positive; 19.7% stale-authority distractors; $\kappa = 0.71$). Cards were authored by the legal-research annotators from public product documentation and complaint allegations under a fixed guideline. Pairs are stratified by product archetype and year so that every archetype appears in train, validation, and test (1,112 / 372 / 372 pairs); the test split is not held-out by unseen archetype. Few-shot exemplars and the entailment threshold use only train/validation; *all reported metrics use the 372 held-out test pairs*. Table 7: VIGIL Full reaches 0.78 F1 (+0.06 vs. no-gate; +0.05–0.13 vs. external baselines). VIGIL without gating (0.72) sits slightly below RAG-only (0.73), indicating that the agentic loop without a validity filter can reinforce stale but lexically close precedent; the gate is the component that improves PLRE. Gains partly reflect rejection of stale distractors (19.7% of pairs). We report the full held-out set only (no non-distractor-only F1; Limitations) and treat deltas as point estimates under a frozen protocol.

5.6 RQ5: Evidence-Coverage Risk Control and Temporal Shift

Calibrating the evidence threshold to $\alpha \in \{0.05, 0.10, 0.20\}$ yields omission risk 0.043/0.082/0.171 in random splits. Under a temporal split (2023–24 → 2025) the achieved risk rises to 0.061/0.113/0.196, exceeding the target at α=0.05 and 0.10 (Table A.9), which motivates rolling re-calibration. To probe dependence on

GPT-4o, an open-weights Llama-3.1-70B+bge stack runs unchanged, with full-test-set macro F1 0.81→0.76 and 2.4 vs. 2.6 iterations (Table A.10). Baselines were not re-run on this stack, so this does not show a backbone-independent advantage.

6 Related Work

Legal NLP and entailment. Legal NLP has studied judgment prediction (Chalkidis et al., 2019), statutory reasoning (Holzenberger et al., 2020), legal benchmarks (Chalkidis et al., 2022; Guha et al., 2023), case retrieval and entailment (Rabelo et al., 2022), and summarization (Huang et al., 2021; Manor and Li, 2019), often with domain-adapted encoders (Chalkidis et al., 2020; Zheng et al., 2021; Xiao et al., 2021). Audits show that LLMs and commercial legal RAG tools still hallucinate (Dahl et al., 2024; Magesh et al., 2025). Prior work also studies overruling detection as a classification task over case text (Zheng et al., 2021; Guha et al., 2023); we instead extract treatment edges to build time-indexed validity intervals and apply them as a retrieval gate. These lines of work take legal text as the premise, treat the corpus as static, and do not test whether retrieved authority remained operative at a given date. PLRE changes the inference unit: product attributes are mapped to legal-risk theories in a directional, product-conditioned, and temporal manner.

Time-sensitive QA, citators, and temporal graphs. Time-sensitive QA conditions answers on the moment a question is asked (Zhang and Choi, 2021; Chen et al., 2021; Liška et al., 2022; Kasai et al., 2023; Dhingra et al., 2022); temporal IR uses document time as a ranking signal (Campos et al., 2014). Both treat time as a property of facts or queries. In PLRE, time enters through supersession: a later document alters the evidential status of an earlier one. Commercial citators track negative treatment (LexisNexis, 2024; Thomson Reuters, 2024); citation-network analysis and edge labeling characterize the underlying graph (Fowler et al., 2007; Sadeghian et al., 2018). VIGIL attaches validity intervals to extracted treatment edges and applies them as a retrieval-time gate (Eq. (3), Fig. 1).

RAG, agents, and risk control. RAG (Lewis et al., 2020; Karpukhin et al., 2020; Gao et al., 2023) typically executes a fixed retrieval plan before generation; work on knowledge conflicts and outdated parametric knowledge shows that generators may prefer stale but familiar evidence unless retrieval is constrained (Longpre et al., 2021; Chen et al., 2022; Xie et al., 2024). Agent frameworks (Yao et al., 2023; Shinn et al., 2023; Wei et al., 2022; LangChain, Inc., 2024; Significant Gravitas, 2023; Park et al., 2023; Wang et al., 2024a,b) benefit from iterative tool use yet stop when the model itself deems its answer satisfactory. VIGIL ties stopping to an externally inspectable predicate over evidence support and mitigation coverage, and employs conformal risk control (Angelopoulos et al., 2024, 2025; Geifman and El-Yaniv, 2017) narrowly for evidence omission rather than legal correctness.

7 Discussion

The central result is that a validated treatment layer, applied as a hard retrieval gate, changes PLRE behavior: on the held-out set, stale citations fall from 12.7% to 4.1% and F1 rises by 0.06 relative to the identical pipeline without gating (Table 5). Independent checks place negative-treatment micro-F1 at 0.80 (gold spans) and 0.81 (commercial citator; Table 1). These figures are point estimates; we treat them as evidence for the mechanism, not as significance-tested superiority.

Two controls bound the claim. First, without the gate, the agentic loop alone does not help on PLRE (0.72 vs. RAG-only 0.73), so gains are not explained by extra test-time compute. Second, a surface-form heuristic recovers many explicit OVERRULES/SUPERSEDES mentions but almost no material narrowing (Appendix D), and citator-oracle intervals flip only 3.8% of held-out pairs—the extracted layer already tracks the commercial reference for downstream decisions. We do not evaluate B5, pure date filters, or learned temporal re-rankers on PLRE; a date filter cannot recover supersession end points from later opinions, which is the extraction problem this work targets.

On severity triage (n=82), VIGIL leads iteration-matched B5 by a small margin (0.79 vs. 0.75); that comparison is secondary to PLRE. Remaining limits—uncertainty quantification, non-distractor breakdowns, dispute-blocked splits, and broader baselines—are listed after the Conclusion.

8 Conclusion

We introduced PLRE and VIGIL, whose extracted treatment layer gates retrieval by operative authority. On held-out pairs, gating lifts PLRE F1 by 0.06 and cuts stale citations from 12.7% to 4.1% rela-

tive to flat similarity. The layer reaches negative-treatment micro-F1 0.80 on gold spans and 0.81 against a commercial citator.

Limitations

What this paper does and does not claim. Our primary claim is mechanistic: an *extracted* treatment layer, validated against gold spans and a commercial citator, can be applied as a hard validity gate at retrieval time, and doing so changes PLRE outcomes and stale-citation rates relative to the same pipeline without the gate (Table 5). We report point estimates and do not claim statistical significance for any comparison. The contribution is the validated gate construction and its downstream effect under matched retrieval, not a leaderboard win under every possible baseline.

Statistical uncertainty. Per-instance prediction vectors for paired bootstrap on the main tables were not retained. We therefore present point estimates only and release evaluation scripts and seeds so that independent re-runs can attach CIs. Aggregate mean$\pm$std over five seeds (Appendix) are run-level summaries, not a substitute for case-paired tests. On triage (n=82), the VIGIL–B5 gap (0.79 vs. 0.75) is modest and is not offered as a significant improvement. On PLRE (n=372 pairs), the $+0.06$ F1 and 12.7%$\rightarrow$4.1% stale-cite reductions are the operational magnitudes of interest; without CIs they should be read as directional under the frozen protocol.

Baselines not evaluated on PLRE. B5 (iteration-matched self-refinement), token-matched refinement, pure document-date filters, and learned temporal re-rankers were not run on PLRE. A date filter cannot encode supersession by *later* opinions (the end point e_v is not a property of document v), so it is not a substitute for treatment extraction; we leave a head-to-head re-ranker comparison to follow-on work. B5 on severity labels is a triage control only.

Distractor composition. Stale-authority distractors are 19.7% of PLRE pairs. We do not report a non-distractor-only F1. Table 5 already includes the hard gate vs. no-gate ablation on the full held-out set; isolating the non-distractor subset would further test ranking among still-open authorities but is not required to establish that the gate removes closed support.

Split structure. Case and PLRE splits are stratified by year and type (or archetype), not by dispute identity. Related dockets of one dispute can cross the train/test boundary. Primary PLRE metrics use held-out pairs only, with few-shot exemplars and thresholds taken from train/validation, but a dispute-blocked re-split remains desirable.

B1, extractors, transfer, and raters. B1 uses additive scores only; retained pre-labels used composite bonuses, so B1 is not scored against its own labels. Negative-treatment recall is 0.77 on gold spans (citator missed-action rate 20.4%). Protocol P1 is incomplete. Annotators and raters were institutional staff. Conformal coverage weakens under temporal shift. The dataset is U.S.-federal and skewed toward large defendants (top three: 41.3% of cases).

Reproducibility. Primary runs used OpenAI API snapshot `gpt-4o-2024-08-06` (June 2025). Azure OpenAI retired that version string on 2026-02-27; OpenAI API access is not guaranteed. Bit-identical re-runs require that snapshot.

Future work. Larger adjudication, paired bootstrap CIs, non-distractor and dispute-blocked audits, B5/token-matched and temporal re-ranker controls on PLRE, and broader coverage.

Ethical Considerations

Decision-support boundary. VIGIL emits risk intelligence, evidence summaries, and mitigation candidates with provenance and residual uncertainty for qualified human reviewers. It is not legal advice: it does not replace counsel, predict litigation outcomes, or recommend business decisions, and severity labels are triage proxies that are not interpreted as liability. Every High- or Medium-risk item is presented for review by a qualified person, and unresolved legal-status atoms are surfaced as such rather than resolved automatically.

Human oversight of failure modes. Low-confidence negative-treatment edges enter a human review queue, since missed negative-treatment actions are the failure mode most likely to admit stale authority into a report. Reviewers can distinguish a genuine product–theory match from a retrieval gap or an unresolved legal-status ambiguity.

Data governance. Released data are public docket metadata and machine-generated summaries; we do not redistribute attorney-authored complaint text beyond what is already public, and the P2 citator study releases only the crosswalk, sampling seeds, and agreement statistics, with no redistributed proprietary citator content.

Source records are public U.S. court filings obtained through CourtListener/RECAP and are used for research consistent with that service's terms; news items are represented only by links and derived metadata. Released records contain the names of parties and counsel as they appear in these public filings; we collect no other personal information. VIGIL-Dataset is released under CC BY 4.0 and the code under the MIT license, for research on legal-risk analysis; neither is intended for automated legal decisions. The Protocol P2 comparison used a commercial citator under the authors' institutional research subscription; only agreement statistics and sampling seeds are released, with no redistributed citator content.

Annotators and raters. The three legal research specialists who adjudicated labels and the two specialists who rated reports (§4, §5) are staff of the authors' institution and performed this work within their regular paid duties, without additional compensation beyond ordinary salary. Before annotation began, participants received a written description of the task, the intended research use of their labels, data-handling practices, and the voluntary nature of participation; verbal consent was obtained and recorded in the project log. The work consisted solely of labeling public court documents and collected no personal information about the participants beyond institutional employment status. Under the authors' institutional human-subjects policy for secondary analysis of public records with no identifiable private information about research subjects, the authors determined that the activity does not constitute human-subjects research requiring IRB review (no separate IRB-office determination was sought); this determination is recorded in the released annotation guidelines. Institutional affiliation of annotators is a potential source of bias (Limitations). Annotation guidelines are released with the dataset so that external replications can use independent raters.

Use of AI assistants. AI writing assistants were used to edit and polish the text of this paper. An AI coding assistant also wrote the code for the P1 pilot (Appendix H.1), following a design fixed by the authors; the pilot design was frozen before any run, and its results were reported as produced. Research ideas, system design, and the remaining experiments and analyses are the authors' own.

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Technical Appendix

This appendix collects material that supports reproducibility but exceeds the page budget of the body: dataset inclusion criteria and statistics (App. A), the ingestion-layer scoring matrix (App. B), system architecture, a case study, and the sensing tool (App. C), treatment-edge extraction accuracy (App. D), additional results and ablations (App. E), hyperparameters, prompts, evaluation protocol, and the release plan (App. F), the proof of Proposition 1 with the calibration audit (App. G), and the validation protocols with measured P2 results (App. H).

A Dataset Inclusion Criteria

A.1 Inclusion Criteria and Case Unit

Search. Candidate dockets were retrieved from CourtListener/RECAP for January 2020–December 2025 by searching complaint, amended-complaint, and petition entries for combinations of three keyword families: (i) AI-training terms (e.g., AI training, model training, training data, dataset, LLM); (ii) legal-theory terms (e.g., copyright, DMCA, unauthorized, without permission, pirated, scraping); and (iii) source-specific terms for known data sources (e.g., shadow libraries, video datasets, music lyrics). Legal-news signals were used only to locate dockets, not as cases.

Screening. A reviewer kept a docket only if the complaint puts the development, training, or deployment of an AI system at issue; keyword matches about unrelated software or data disputes were removed. Litigation type (Table A.1) is assigned afterwards from the recorded cause of action, which is why non-copyright categories (trade secret, other) appear.

Case unit. One case corresponds to one docket. Keyword retrieval and manual screening for AI development, training, or deployment at issue yield 1,260 docket records; deduplication of repeated retrievals of the same docket then yields 1,247 cases. Deduplication does not merge related or consolidated dockets, so a single dispute litigated in several dockets contributes several cases. Counts should therefore be read as dockets, not as distinct disputes.

Table A.1 gives annual counts by litigation type after keyword retrieval and manual AI-relevance screening, before docket deduplication (1,260 post-screening records; 1,247 after dedup).

Table A.1: Annual filing breakdown by litigation type. Counts are post-screening filings (1,260) before docket deduplication; the annotated subset contains 1,247 cases.

Year	Copyright	DMCA	Trade Secret	Other	Total
2020	28	9	7	3	47
2021	41	14	10	4	69
2022	89	31	22	8	150
2023	198	72	51	19	340
2024	261	96	64	24	445
2025	122	47	29	11	209
Total	**739**	**269**	**183**	**69**	**1,260**

B Ingestion-Layer Scoring Matrix

Table A.2 lists the additive ingestion-layer scoring matrix (§2) used for real-time triage and as the automated pre-labeler. The critical-composite bonus fires only when copyright and model-training evidence co-occur. This heuristic is a baseline, never ground truth.

Table A.2: Ingestion-layer scoring matrix. The critical-composite bonus fires only when copyright and model-training evidence co-occur.

Evidence Category	Key Indicators	Score
Critical composite	**Copyright ∧ Training co-cited**	**+50**
Copyright direct mention	NOS 820, copyright keyword	+30
Unauthorized collection	scrape, crawl, ingest, harvest	+25
Direct model training mention	train, LLM, transformer, diffusion	+20
Copyright / IP dispute	infringement, DMCA, fair use	+10
Commercial exploitation	profit, revenue, subscription	+10
Class action filing	class action, putative class	+5
Licensed data cooperation	contract, licensing, partnership	−10

C Open-Source Implementation Details

C.1 System Architecture and Discovery

Figure A.1 shows how the sensing pipeline feeds the agentic layer evaluated in the main text. VIGIL's discovery layer ingests CourtListener/RECAP (Free Law Project, 2023), docket-backed legal-media signals, and keyword-filtered news. A two-stage deduplication module first matches docket IDs and titles, then optionally removes Gemini-embedding neighbors above cosine 0.85 (Algorithm 1). Detected filings are processed by combining docket metadata, NER over complaint

text, regex extraction for damages, and LLM summarization for the narrative fields. All code will be released, as described in Appendix F.

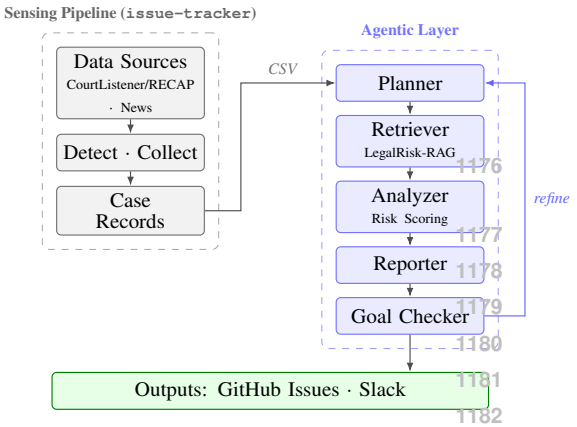


Figure A.1: VIGIL architecture. The sensing pipeline (left) detects filings, collects documents, and prepares case records; the agentic layer (right) plans, retrieves time-valid precedent, scores risk, and reports, with the Goal Checker driving a refine loop until evidence coverage is met.

C.2 Case Study and Live Monitoring

Case studies illustrate mechanism behavior, not outcome prediction. In a held-out PLRE query about output-CMI risk, flat RAG ranked a lexically close authority whose interval had already closed; validity gating removed it from support and promoted a still-operative authority, preventing a stale-citation report. In 30 days, `issue-tracker` processed 1,159 items, surfaced 94 unique filings after 92% deduplication, and reached discovery-layer Precision 0.90/Recall 0.84.

C.3 Sensing Tool

The `issue-tracker` tool runs on an adaptive, configurable CI automation schedule: hourly during a configurable business-hours window and two-hourly off-hours. On-demand execution via `workflow_dispatch` is also supported. Its principal modules are:

- **Query engine**: five CourtListener queries (complaint/petition filters with AI keywords, REST API v4) plus four news queries over Google News RSS.
- **Ingestion pipeline**: orchestrates polling, RSS monitoring, docket backtracking, and cross-issue deduplication over a configurable window.
- **Complaint parser**: extracts cause-of-action language, training snippets, and parties from RECAP PDFs.
- **Risk scorer**: applies the seven-category additive matrix *plus* the critical-composite (+50) co-occurrence rule of Table A.2 to produce a 0–100 score with a four-band severity indicator.
- **Hybrid deduplication**: a two-stage pipeline (Algorithm 1) that combines lexical title/docket matching with optional semantic similarity using Gemini `text-embedding-004` embeddings (cosine threshold 0.85), enabled by a configuration flag.
- **Reporting**: optional day-close summary generation (provider-agnostic LLM back-end).
- **Reporting integrations**: posts to GitHub Issues with a three-tier hierarchical structure (issue body / morning trend summary / evening deduplicated table) and a parallel Slack notification.

Algorithm 1 Hybrid Lexical–Semantic Deduplication

Require: Candidate items $\mathcal{N}_{\text{new}}$, baseline items $\mathcal{N}_{\text{base}}$ (prior issues), threshold σ
Ensure: Deduplicated set $\mathcal{N}^*$

$T_b \leftarrow \textsc{ExtractTitles}(\mathcal{N}_{\text{base}})$
$\mathcal{R} \leftarrow \{i \in \mathcal{N}_{\text{new}} \mid \text{title}(i) \notin T_b\}$ {Stage 1: exact-title lexical filter}
if $\textsc{LLMEnabled} \wedge |\mathcal{R}| > 0$ **then**
  $E_{\text{curr}} \leftarrow \textsc{GeminiEmbed}(\{\text{title}(i) : i \in \mathcal{R}\})$
  $E_{\text{base}} \leftarrow \textsc{GeminiEmbed}(T_b)$
  $\mathcal{S} \leftarrow \{i \in \mathcal{R} \mid \exists e \in E_{\text{base}} : \cos(E_{\text{curr}}[i], e) \geq \sigma\}$
  $\mathcal{R} \leftarrow \mathcal{R} \setminus \mathcal{S}$ {Stage 2: semantic filter}
end if
return $\mathcal{N}^* \leftarrow \mathcal{R}$ {≈2 embedding calls/run; free-tier compatible}

Table A.3 lists the key configuration parameters of `issue-tracker`, covering API credentials, scheduling, deduplication, and reporting options. All secrets are injected via GitHub Secrets; environment variables are set via GitHub Variables.

Table A.3: Key configuration parameters of `issue-tracker` (identifiers anonymized for review). Secrets via GitHub Secrets; variables via GitHub Variables.

Parameter	Default	Description
Lookback window	3 days	Days of history to retrieve
CourtListener token	—	CourtListener API v4 token
Issue title base	—	Base title for GitHub Issues
Issue label	(project label)	GitHub Issue label
LLM API key	—	LLM API key (optional)
LLM model	`gemini-flash-latest`	Gemini model id
Trend window	(unset)	Trend-summary window (days)
Semantic dedup	off	Enable semantic dedup (stage 2)
Semantic dedup threshold	0.85	Cosine threshold for semantic dedup
Daily visual brief	off	Enable a text-to-image model daily visual brief
Cross-issue dedup window	(unset)	Cross-issue dedup window (days)
Show docket candidates	off	Show uncertain candidates
Collapse long cells	off	Fold long table cells
Collapse article URLs	off	Collapse article URL lists
Debug	off	Verbose logging

D Treatment-Edge Extraction Accuracy

Table A.4 reports the per-relation accuracy of the treatment-edge extractor (§3.4) against the hand-labeled gold span set. Negative-treatment edges drive validity-interval closure (SUPERSEDES/OVERRULES) or down-weighting (material NARROWS) and are reported separately from the high-frequency CITES relation, with a micro-average over the three negative-treatment relations.

Table A.4: Treatment-edge extraction accuracy against the 600 hand-labeled gold citation spans (§3.4).

Relation	Precision	Recall	F1
CITES	0.94	0.96	0.95
DISTINGUISHES	0.81	0.76	0.78
NARROWS (material)	0.79	0.71	0.75
SUPERSEDES	0.85	0.78	0.81
OVERRULES	0.88	0.82	0.85
Negative-treatment (micro)	0.84	0.77	0.80

Regex heuristic baseline. As a simpler alternative to the LLM edge classifier, a surface-form heuristic flags citation sentences containing keywords such as `overruled`, `superseded`, `abrogated`, `overturned`, or `no longer good law`. On the same 600 gold spans, negative-treatment micro precision/recall/F1 are 0.71 / 0.41 / 0.52 (vs. LLM 0.84 / 0.77 / 0.80). The heuristic recovers many OVERRULES/SUPERSEDES mentions but almost no material-NARROWS and cannot label DISTINGUISHES polarity. We retain the LLM classifier and route low-confidence closures to human review.

E Additional Results and Ablations

Table A.5: Severity-label agreement on the full held-out test set, including retained pre-labels (RQ1; point estimates, reported for scale). B5 is matched to VIGIL's mean iteration budget (not to total LLM calls or tokens).

System	F1	Precision	Recall
B1 Additive-only	0.62	0.58	0.67
B2 Single-Pass LLM	0.72	0.74	0.70
B3 RAG-only	0.74	0.75	0.72
B4 Single-Agent	0.76	0.77	0.75
B5 Self-Refinement	0.77	0.78	0.76
VIGIL Full	**0.81**	**0.82**	**0.80**

Table A.5 reports severity-label agreement on the full held-out test set (RQ1); primary claims use the human-adjudicated subset (Table 4).

Table A.6 reports the component-wise ablation of the agentic loop, showing the contribution of the Goal Checker, Planner, and Retriever query-expansion to risk classification F1 and retrieval Recall@10 (RQ3 in the main text).

Table A.6: Agentic-loop ablation: component-wise contribution to risk classification F1 and retrieval Recall@10 (mean over 5 evaluation runs). Referenced from the main text (RQ3).

Variant	F1	Recall@10
VIGIL (Full)	**0.81**	**0.84**
− Goal Checker (Single-Agent)	0.76	0.84
− Planner (fixed task order)	0.78	0.84
− Retriever query-expansion	0.79	0.76

Table A.7 reports Recall@10 on a held-out development set of 200 query–document pairs for three retriever configurations. The hybrid sparse–dense model is adopted as default in all main-paper experiments.

Table A.7: Retriever ablation. Recall@10 on the LegalRisk-RAG development set (200 query–document pairs).

Retriever Configuration	Recall@10
Dense (`text-embedding-3-large`)	0.79
LegalBERT (fine-tuned on RECAP)	0.81
Hybrid (BM25 + neural re-ranking)	**0.84**

F Reproducibility: Hyperparameters, Prompts, Protocol

F.1 Goal-Conditioned Agentic Loop (Full Procedure)

Algorithm 2 gives the full VIGIL agentic loop summarized in the main text.

Algorithm 2 VIGIL Goal-Conditioned Agentic Loop

Require: Document corpus $\mathcal{D}$, product spec P, goal predicate $\mathcal{G}$, evidence threshold (k, τ), max iterations $T_{\max}$
Ensure: Verified risk report R^*, mitigation guidance G^*

$R_0 \leftarrow \emptyset$; $t \leftarrow 0$
while $\neg\, \mathcal{G}(R_t) \ \wedge\ t < T_{\max}$ **do**
  $\mathcal{T}_t \leftarrow \textsc{Planner}(R_t, \mathcal{G}, P)$ {Decompose unmet sub-goals}
  for each sub-task $tk \in \mathcal{T}_t$ **do**
    $E_{tk} \leftarrow \textsc{Retriever}(tk, \mathcal{D})$ {LegalRisk-RAG}
    if $|E_{tk}| < k \ \vee\ \min \text{score}(E_{tk}) < \tau$ **then**
      $E_{tk} \leftarrow \textsc{Retriever}(\textsc{ExpandQuery}(tk), \mathcal{D})$
    end if
    $C_{tk} \leftarrow \textsc{Analyzer}(E_{tk}, P)$ {Risk score & clustering}
  end for
  $R_{t+1} \leftarrow \textsc{Reporter}(\{C_{tk}\}, R_t)$
  if $\textsc{GoalChecker}(R_{t+1}) = \textsc{True}$ **then**
    break
  end if
  $t \leftarrow t + 1$
end while
$G^* \leftarrow \textsc{Mitigate}(R_t, P)$ {Per-risk action templates}
return $R^* \leftarrow R_t,\ G^*$

F.2 Goal Predicate and Loop Parameters

Table A.8 summarizes the parameters used by Algorithm 2 for all reported experiments.

F.3 Agent Prompt Templates (Schematic)

Each of the four agents (Planner, Retriever, Analyzer, Reporter) is invoked with a structured prompt of the form:

Table A.8: VIGIL goal-predicate and loop hyperparameters used throughout the evaluation.

Parameter	Value	Description
k	3	Min. retrieved documents per risk item
τ	0.62	Min. relevance score (hybrid re-ranker, min-max scaled to $[0, 1]$)
$T_{\max}$	5	Max. agentic loop iterations
θ (LLM)	GPT-4o	Controller/Analyzer T=0.2; edge classifier T=0
Embedding dim.	3072	`text-embedding-3-large`
Mapping threshold	0.72	Bi-encoder product–risk similarity
w: positive / open	1.00	Positively treated or still-open authority
w: cautionary	0.75	Cautionary or questioned treatment
w: material narrows	0.50	Materially narrowed authority
w: overruled/superseded	0.00	Removed by the validity indicator (Eq. (3))

```
[ROLE]: {agent role description}
[INPUTS]: {R_t, P, sub-goal, retrieved evidence}
[GOAL CHECK]: {unmet sub-conditions of G}
[OUTPUT FORMAT]: {structured JSON schema}
```

The full prompt templates, including the few-shot examples of the treatment-edge classifier, are part of the code release (Appendix F, *Code and data release*). The Goal Checker is implemented as deterministic rule evaluation over R_t (no LLM call), making termination decisions auditable.

F.4 Evaluation Protocol Notes

Dataset split. The 70/10/20 train/validation/test split is stratified by year and litigation type and held fixed across all baselines (B1–B5 and VIGIL (Full)). The 5 random seeds vary controller/Analyzer decoding (T=0.2) and few-shot exemplar order; the retrieval index, treatment graph, edge classifier (T=0), and held-out test membership remain fixed.

Likert evaluation. Two legal specialists, blind to system identity, rated 75 reports (the same 15 cases for each of the five conditions, so comparisons are paired by case) on a 5-point scale across five dimensions. Reports were presented in randomized order; system identifiers were stripped.

Statistical reporting. Main-text tables report point estimates. Per-instance prediction vectors for paired bootstrap on the primary tables were not retained, so we do not report case-level CIs or paired tests there; scripts and seeds are released for reruns. Multi-run means in Tables A.6 and A.10 are aggregate run-level summaries over five seeds (controller/Analyzer decoding at T=0.2 and few-shot exemplar order; fixed retrieval index, treatment

graph, edge classifier at T=0, and test membership). Those summaries are not a substitute for per-instance logs. Likert differences in RQ2 use the Wilcoxon signed-rank test on per-case mean ratings without multiplicity correction. Inter-rater κ between the two raters is 0.68/0.71/0.65/0.74/0.62 across the five scales (mean 0.68).

Code and data release. We will release all code used for the reported results: the sensing pipeline (`issue-tracker`), treatment-edge extraction and graph construction, the PLRE annotation guidelines and evaluation scripts, and the agentic loop, together with the treatment-weight schedule $w(\cdot)$, prompt templates with few-shot examples, split files, seeds, VIGIL-Dataset (CC BY 4.0; code under MIT), and the SCOTUS supersession testbed of Appendix H.1, which is built only from public, licence-free sources. For review, an anonymized copy of the code and data is provided as supplementary material; the full public release will follow the review period.

Models, compute, and software. The controller and Analyzer use GPT-4o at temperature 0.2; the edge classifier uses the same snapshot at temperature 0 (snapshot `gpt-4o-2024-08-06`; primary evaluation runs: June 2025); embeddings use `text-embedding-3-large` and, for deduplication only, Gemini `text-embedding-004`. The robustness re-run (Table A.10) uses Llama-3.1-70B (70B parameters) with `bge-large`, and the retriever ablation uses LegalBERT fine-tuned on RECAP text. As an indication of scale, each case requires on average 2.4 loop iterations (at most $T_{\max}$=5), and the test-set evaluations use five runs (seeds as described above). Exact software packages and versions (BM25, re-ranker, bootstrap and conformal-calibration code) are pinned in the dependency file of the released code.

G Proof of Proposition 1 and Calibration Protocol

Table A.9 reports evidence-omission risk under random and temporal calibration splits. calibrated thresholds meet the budget α on random splits, while the temporal split exceeds it at $\alpha \in \{0.05, 0.10\}$, showing that exchangeability is stressed under temporal shift (RQ5). Table A.10 reports full-system macro F1 and mean loop iterations when the GPT-4o backbone is replaced by an open-weights stack. Only VIGIL was re-run on the open-weights stack; baselines were not, so the table does not support cross-backbone comparisons with the baselines.

Table A.9: Evidence-omission risk under random and temporal calibration splits. Lower is better; values should be read as evidence-coverage risk, not severity-label correctness.

Split	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$
Random resplits	0.043	0.082	0.171
Temporal 2023–24 → 2025	0.061	0.113	0.196

Table A.10: Full VIGIL system under two backbones (mean ± std over runs). Baselines were not re-run on the open-weights stack.

Configuration	**Macro F1**	**Mean iters**
GPT-4o + `text-embedding-3-large`	0.81 ± 0.02	2.4 ± 0.3
Llama-3.1-70B + `bge-large`	0.76 ± 0.02	2.6 ± 0.3

G.1 Termination Property (Proof)

Let $\Phi(R_t) \in \mathbb{Z}_{\geq 0}$ count the unmet atomic sub-conditions of $\mathcal{G}$ (one per missing severity label, per under-supported evidence requirement, and per missing mitigation) over the risk items in R_t. By assumption (A1) *monotone retrieval*, the evidence set attached to any item is non-decreasing across iterations, so a sub-condition once satisfied by $\mathcal{G}_{\text{evidence}}$ or $\mathcal{G}_{\text{guide}}$ remains satisfied; hence no satisfied atom can become unmet and $\Phi(R_{t+1}) \leq \Phi(R_t)$. By assumption (A2) *attainable progress*, whenever $\Phi(R_t) > 0$ and at least one unmet atom is *attainable* (the required evidence exists in $\mathcal{D}$), the Planner schedules it and the Retriever/Analyzer resolves it, giving $\Phi(R_{t+1}) \leq \Phi(R_t) - 1$. Therefore, if every atom is attainable, Φ strictly decreases each iteration until $\Phi = 0$, i.e. $\mathcal{G} = \textsc{True}$, in at most $\Phi(R_0)$ steps. If some atom is unattainable, Φ cannot reach 0; the loop runs to $T_{\max}$ and the Goal Checker reports the residual $\Phi(R_{T_{\max}})$ unmet atoms, which are exactly the items requiring human escalation. □

Remark. (A1) holds by construction because `EXPANDQUERY` unions new hits with the prior evidence cache; (A2) is the design assumption that one tractable sub-goal is closed per pass. We do not verify it directly; the mean iteration count (2.4, below $T_{\max}$=5) is consistent with it but does not establish it.

G.2 Conformal Risk-Controlled Termination

We calibrate the evidence threshold τ (and, jointly, k) so that termination meets a user-specified omission-risk budget α. For calibration cases

c_i, $i = 1, \ldots, n$, let the bounded loss $L_i(\tau)$ indicate that a retrievable, human-labeled High-risk theory of c_i is omitted from $R^*(c_i)$, and let $\hat{R}_n(\tau) = \frac{1}{n}\sum_i L_i(\tau)$, which is non-increasing as τ decreases (more evidence admitted). Following conformal risk control (Angelopoulos et al., 2024), choosing $\hat{\tau} = \sup\{\tau : \frac{n}{n+1}\hat{R}_n(\tau) + \frac{1}{n+1} \leq \alpha\}$ (the strictest threshold that still meets the budget; the risk bound is non-decreasing in τ) controls $\mathbb{E}[L_{n+1}(\hat{\tau})] \leq \alpha$ for an exchangeable test case under the calibration assumptions. For the joint (k, τ) we use a Learn-then-Test grid (Angelopoulos et al., 2025), treating each configuration as a hypothesis and controlling the family-wise error via fixed-sequence testing; this gives the high-probability guarantee $\Pr[R(\hat{k}, \hat{\tau}) \leq \alpha] \geq 1 - \delta$ rather than the expected-risk bound above. The achieved-vs-target risk is reported in RQ5 and Table A.9.

G.3 Temporal-Split Evidence-Coverage Audit

Table A.9 separates the standard random calibration/test split from a stricter temporal split. The temporal split does not invalidate the calibrated procedure; it shows that the exchangeability assumption is empirically stressed by evolving litigation streams and motivates rolling or weighted calibration in deployment.

H Pre-Specified Validation Protocols

The two construct-validity steps flagged in the Discussion—transfer of the validity-gating modifier to an external retrieval stack, and agreement of the automatically extracted treatment graph with an authoritative citator—are specified here as *pre-specified protocols*. Protocol P1 (external-stack transfer) is reported as a protocol only and is not yet executed; Protocol P2 (citator agreement) has been executed and its measured results are reported in Appendix H.2. The protocols fix the datasets, conditions, metrics, statistical tests, and acceptance criteria *before* execution so that the outcome cannot be tuned post hoc. Both protocols are designed to be runnable on public resources plus a single licensed citator account, and both release their analysis scripts, query/seed manifests, and (for the citator study) the category crosswalk rather than any redistributed proprietary content.

H.1 Protocol P1: External-Stack Transfer of the Validity-Gating Modifier

Hypothesis. The supersession-aware score of Eq. (3)—a base similarity multiplied by a time-validity indicator $\mathbf{1}[s_v \leq t_q < e_v]$ and a treatment weight $w(\text{treat}(\cdot))$—improves *external* legal retrievers that do not use VIGIL's agentic loop, dataset, or goal predicate, which would indicate that the mechanism transfers rather than being VIGIL-specific.

Host retrievers (unmodified, external). (i) BM25; (ii) a general dense retriever (`text-embedding-3-large` or an open `bge`/`e5` encoder); (iii) a legal encoder (Legal-BERT). Each is the "host" stack; the modifier is applied only as a post-hoc re-scoring layer ($\text{score}' = \text{score}_{\text{host}} \cdot \mathbf{1}[\cdot] \cdot w(\cdot)$), with *no* re-indexing and *no* VIGIL loop.

Evaluation corpora (disjoint from VIGIL-Dataset). (a) a COLIEE-style public case-retrieval benchmark; and (b) a temporally held-out CourtListener slice whose jurisdiction/time window does not overlap the VIGIL-Dataset window. Setting (b) is the decisive transfer test because it contains time-varying authority outside VIGIL's training window.

Conditions. For each host retriever: {ungated baseline} vs. {+ indicator only} vs. {+ indicator and treatment weight}, mirroring the decomposition of Table 5. Each query carries a query date t_q; an authority closed before t_q is treated as stale (non-relevant) in the gold judgments.

Treatment-label source (confound control). To avoid inheriting VIGIL's extractor noise, the external test uses treatment labels from a source independent of VIGIL's extractor (gold/citator-derived). We report two variants per condition—*gold-treatment* and *extractor-treatment*—so the transfer claim is separated from extraction quality.

Metrics. Retrieval quality (nDCG@10, Recall@10, MRR) and a *stale-citation rate* (fraction of top-k closed before t_q). The primary endpoints are the paired deltas (gated − ungated) per host retriever.

Statistics. Paired bootstrap over queries (10,000 resamples) with 95% CIs on each delta; Holm correction across the three host retrievers; a pre-specified non-inferiority margin ($\delta_{\text{NI}} = 0.5$ nDCG points) requiring the modifier not to hurt on queries with no negative-treatment in scope.

Acceptance criterion (pre-specified). P1 succeeds iff the modifier yields a positive nDCG@10 delta *and* a reduced stale-citation rate on at least two of three host retrievers, the effect persists on the temporally held-out slice (b), and non-inferiority holds on the no-stale subset. Failure

on slice (b) but success on (a) is reported as "in-distribution only," which would *not* support the transfer claim.

P1 pilot (gold treatment, slice-(b) analogue). P1 itself has not been executed. We ran a pilot on public data whose design, including the acceptance criterion below, was frozen before any retrieval run. It deviates from P1 as follows: no slice (a); no extractor-treatment variant; a cluster bootstrap by gold pair instead of a per-query bootstrap. All deviations are listed in the released pilot code. Gold closures come from the Constitution Annotated table of overruled U.S. Supreme Court decisions (Congressional Research Service, 2022), matched to Caselaw Access Project opinions (Harvard Law School Library, 2024). Overrulings decided in or after 2020 are excluded, so the pilot is disjoint from the VIGIL-Dataset window. This leaves 253 pairs and a corpus of 29,030 opinions. Queries are citing passages from later opinions with case names, citations and years masked:

- *post-closure* queries cite the overruling case B, so the overruled case A is stale (984 queries in 208 pairs);
- *pre-closure* queries cite A before its first overruling (1,061 queries);
- *control* queries cite cases absent from the table (1,133 queries).

Each host retrieves the top 100 among opinions decided before the query date. The hosts are BM25 (Robertson and Zaragoza, 2009), `bge-small-en-v1.5` (Xiao et al., 2024) and LegalBERT (Chalkidis et al., 2020), the dense hosts truncated for CPU. The pre-fixed criterion requires significant (Holm) nDCG@10 gains and stale@10 reductions on at least two of three hosts, with non-inferiority on controls. Table A.11 reports the results.

The pilot *does not support* transfer under this criterion: only one of three hosts (bge) meets it. With gold intervals the gate removes every overruled case from the top 10 on all hosts. Non-inferiority holds everywhere, and on pre-closure queries the gate never removed the gold case. However, nDCG@10 gains are at most 0.13 points, because removal helps only when the stale case outranks the correct one. LegalBERT is a weak host here (nDCG@10 2.3) and almost never retrieves the stale case, so it tests the gate little. On partial overrulings, $w \in \{0.25, 0.5, 0.75\}$ yields identical rankings, so the pilot is uninformative about $w(\cdot)$.

Table A.11: P1 pilot, gold-treatment variant: indicator gate vs. ungated host on post-closure queries of full overrulings ($n = 984$, 208 pair clusters). *Stale@10*: queries with the overruled case in the top 10 (ungated). Δ: gated $-$ ungated with 95% cluster-bootstrap CIs (10,000 resamples); *Holm-adjusted $p < 0.05$ across hosts. *Ctrl*: ΔnDCG@10 lower bound on control queries (non-inferiority margin -0.5).

Host	nDCG@10	Stale@10	ΔnDCG@10	ΔStale@10 (pts)	Ctrl LB	Met
BM25	28.88	8.3%	$+0.09$ $[-0.58, +0.52]$	-8.3^* $[-10.7, -6.2]$	$+0.04$	no
bge-small	8.35	5.5%	$+0.13^*$ $[+0.04, +0.27]$	-5.5^* $[-7.1, -3.9]$	$+0.00$	yes
LegalBERT	2.30	0.4%	$+0.00$ $[+0.00, +0.00]$	-0.4^* $[-0.8, -0.1]$	$+0.00$	no

In a post-hoc sensitivity analysis (not pre-specified), we exclude four BM25 queries whose gold case was itself later overruled (*United States v. Rabinowitz*, overruled in 1969). BM25's nDCG@10 delta then becomes $+0.37$ $[+0.19, +0.57]$, and two hosts would meet the criterion. Because this exclusion was chosen after seeing results, the pre-specified verdict stands.

Released testbed. The pilot's queries, qrels, gold closures, corpus identifiers and a scorer that accepts any TREC run are released as a testbed for supersession-aware retrieval, together with scripts that rebuild the corpus from the public sources. The four queries with an already-overruled gold case are flagged. The testbed supports work this paper leaves open: an extractor-treatment variant (replace the gold closures with predicted ones), stronger full-document hosts, soft versus hard gating, and the effect of gating on which authority a generator cites.

H.2 Protocol P2: Citator Agreement for the Treatment Graph

Objective. Replace "validated against our own gold spans" with "validated against an authoritative citator" for the negative-treatment edges (SUPERSEDES, OVERRULES, material NARROWS) that drive interval closure or down-weighting.

Sampling frame and power. A stratified random sample of edges from the current graph (1,812 nodes, 4,760 edges), stratified by relation type and polarity and oversampling negative-treatment edges. Target $n = 400$ edges with ≥ 150 negative-treatment edges; this supports a 95% CI half-width of ≤ 0.07 on Cohen's κ at an anticipated $\kappa \approx 0.75$. The exact draw is fixed by a released seed before any citator lookup.

Reference standard. For each sampled (citing $\rightarrow$ cited) pair and date, retrieve the citator treatment signal (e.g., KeyCite/Shepard's negative, cau-

tionary, or positive flags) and map it to VIGIL's relation/polarity taxonomy through a *pre-specified, published crosswalk*. Two annotators perform this mapping independently and blind to VIGIL's predicted edge; a third adjudicates disagreements. Inter-annotator agreement on the citator mapping itself is reported (Cohen's κ).

Primary endpoints.

- *Edge agreement*: precision/recall/F1 of VIGIL edges vs. citator-derived edges, per relation and micro-averaged over negative treatment.
- *Interval-closure or down-weighting agreement*: whether VIGIL acts (closes the interval for OVERRULES/SUPERSEDES, down-weights for material NARROWS) no later than the citator's negative-treatment date; the action-date error distribution (in days); and the rates of *missed actions* (citator negative, VIGIL neither closes nor down-weights) and *false actions* (VIGIL acts, citator still good).
- *Point-in-time correctness*: validity intervals are checked at multiple historical dates, not only the latest status, because citators are themselves time-varying.

Downstream link. On the disagreement subset, the PLRE/stale-cite audit is re-run with citator-corrected closures to measure the *decision-flip rate*, connecting construct validity to risk-decision outcomes.

Acceptance criterion (pre-specified). Before sampling, we froze two thresholds in the released protocol note: negative-treatment micro-F1 ≥ 0.80 vs. the citator, and missed-action rate $\leq$ the internal-gold miss rate measured for the same extractor configuration on the 600-span gold set ($\approx$23%). The second threshold is deliberately matched to the extractor rather than to a citator-oracle upper bound; it tests whether citator disagreement exceeds the extractor's own gold-set error, not whether the extractor is perfect. Relation types failing the bar are routed to human review.

Licensing and ethics. Citator outputs are proprietary and are used solely as an evaluation reference under a licensed account. The release contains the crosswalk, sampling seeds, and agreement statistics, but no redistributed citator content, consistent with the data-governance boundary in the main text.

Measured P2 results. The audit sampled $n = 400$ treatment candidates, including 162 negative-treatment candidates. Inter-annotator agreement on the citator-to-VIGIL crosswalk was substantial ($\kappa = 0.78$). Edge-level agreement was measured against the citator-derived reference, and interval-closure or down-weighting behavior plus downstream impact were reported. We report micro-F1 rather than Cohen's κ as the primary agreement metric because the relation inventory is multi-class and imbalanced; pairwise κ on the negative-treatment vs. other dichotomy is 0.74 on the same sample. Negative-treatment micro-F1 is 0.81 and the missed-action rate (missed closures for OVERRULES/SUPERSEDES or missed down-weighting for material NARROWS) is 20.4%, satisfying the pre-specified P2 threshold and staying below the $\approx$23% internal-gold miss rate of the extractor configuration evaluated in Table 6, so the measured downstream effect there is not optimistic relative to the citator reference. Material NARROWS remains the weakest relation and is retained in the human-review queue. Edge-level agreement is reported in Table A.12; interval-closure agreement and downstream decision impact are reported in Table A.13. Action-level TP equals relation-level micro TP (129). Predicted positives sum to 156 by inverting per-relation precision; of the 27 relation-level false positives ($156-129$), only 7 change the downstream closing/down-weighting *action* relative to the citator (false-action rate $7/156$); the remainder are relation-type confusions that still trigger the same action class.

Table A.12: Measured P2 edge agreement vs. the citator-derived reference. n is the number of citator-positive instances for that relation (reference positives). Precision uses predicted positives; recall uses n as the denominator. Counts are integers (e.g., overrules recall $45/54$). Micro-average pools the three negative-treatment relations.

Relation	n	Precision	Recall	F1
OVERRULES	54	0.87	0.83	0.85
SUPERSEDES	47	0.82	0.79	0.80
material NARROWS	61	0.80	0.77	0.78
Negative-treatment (micro)	162	**0.83**	**0.80**	**0.81**
Cautionary treatment	78	0.81	0.74	0.77
Positive/neutral citation	160	0.93	0.95	0.94

Table A.13: Measured P2 action agreement (interval closure for OVERRULES/SUPERSEDES, down-weighting for material NARROWS) and downstream decision impact against the citator-derived reference.

Metric	Value
Matched neg.-treatment actions (relation-level TP; same count as micro edge TP)	129 / 162
Missed-action rate	20.4%
False-action rate (7/156 predicted neg.-treatment actions)	4.5%
Median action-date error	18 days
Mean action-date error	31 days
90th-percentile action-date error	74 days
Point-in-time validity accuracy	0.88
PLRE decision-flip rate after citator correction (% of all held-out PLRE pairs)	3.8%